

Comprehensive Plotly Data Visualization Lab

Objective

This extended laboratory session is designed to take students from foundational plotting techniques to advanced, interactive, and multi-dimensional visualizations using Python's Plotly library. Students will utilize a comprehensive dataset containing 200 daily records of business operations.

Dataset Overview

The provided dataset (plotly_lab_data_extended.csv) contains daily metrics. Below is a dictionary of the available fields:

Column Name	Data Type	Description
Date	Datetime	The specific day of the recorded metrics (200 consecutive days).
Region	Categorical	Geographic region (North, South, East, West).
Product_Category	Categorical	The main product line (Electronics, Software, Accessories, Services).
Marketing_Spend	Numeric	Total daily marketing expenditure in USD.
Revenue / Expenses / Profit	Numeric	Financial metrics recorded for the day in USD.
Active_Users / New_Signups	Numeric	User engagement and acquisition metrics.
Customer_Satisfaction	Numeric	Average rating from 1.0 to 5.0.

Column Name	Data Type	Description
Conversion_Rate	Numeric	Ratio of New Signups to Active Users.
Discount_Applied	Categorical	Whether a discount was running on that day (Yes/No).
Month	Categorical	The name of the month, derived from the Date column.

Level 1: Beginner Tasks (Foundations)

1. **Line Chart:** Create a standard line chart showing the trend of Revenue over Date.
2. **Bar Chart:** Aggregate the data to create a bar chart displaying total Revenue by Region.
3. **Scatter Plot:** Plot Marketing_Spend on the x-axis and Revenue on the y-axis to observe potential correlations.
4. **Pie Chart:** Visualize the proportional distribution of records across the different Product_Category types.
5. **Histogram:** Generate a histogram to understand the frequency distribution of Customer_Satisfaction scores. Adjust the number of bins to 15.

Level 2: Intermediate Tasks (Multi-variable Analysis)

6. **Grouped Bar Chart:** Calculate the average Revenue and average Expenses for each Region. Display these side-by-side using a grouped bar chart configuration.
7. **Box Plot:** Create a box plot to compare the distribution of Profit across the four different Product_Category groupings. Note the medians and outliers.
8. **Bubble Chart:** Upgrade your earlier scatter plot. Plot Marketing_Spend vs Revenue, but set the size of the markers based on Active_Users and color them by Region.
9. **Area Chart:** Create an overlapping or stacked area chart showing the progression of both Revenue and Expenses over Date.
10. **Violin Plot:** Use a violin plot to visualize the density and distribution of the Conversion_Rate across different Regions.

Level 3: Advanced Tasks (Complex Interactive

Visuals)

11. **Sunburst Chart:** Create a hierarchical sunburst chart that breaks down the dataset first by Region, then by Product_Category, with the segment sizes determined by total Revenue.
12. **3D Scatter Plot:** Map three numeric variables onto a 3D space: Marketing_Spend (x), Revenue (y), and Profit (z). Use Product_Category for the color mapping.
13. **Density Heatmap:** Generate a 2D density heatmap to visualize the concentration of data points between Active_Users (x-axis) and New_Signups (y-axis).
14. **Facet Grid (Subplots):** Create a grid of line charts. Plot Profit over Date, but facet the charts so that each Region has its own dedicated row (subplot).
15. **Animated Scatter Plot:** Plot Marketing_Spend vs Revenue as a scatter plot. Use the Month column to create an animation_frame, allowing users to hit 'play' and watch how the data clusters change month over month. Color the markers by Discount_Applied.

Deliverables

Students must submit their .ipynb notebook along with a brief markdown summary explaining the business insights gathered from the Advanced Tasks section.